# Year 10 Mathematics

## Graphs 1

3.2 Plotting Straight-Line Graphs

# Plotting Straight-Line Graphs

STARTER · 5 MIN

**Date:** _____ **Title:** Plotting Straight-Line Graphs

**LEARNING OBJECTIVES**

- Complete a table of values for a straight-line graph
- Plot and draw straight-line graphs
- Draw lines such as $x = 3$, $y = -2$ and $x + y = 5$

**STARTER** *Answer in your book. No calculators.*

1. $y = 2x + 1$. Find $y$ when $x = 3$
2. $y = 2x + 1$. Find $y$ when $x = -2$
3. Work out $3 \times (-1) - 4$
4. Solve $2x + 6 = 0$

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. $y = 2 \times 3 + 1 =$ **7**

2. $y = 2 \times (-2) + 1 =$ **$-3$**

3. $-3 - 4 =$ **$-7$**

4. $2x = -6, \quad x =$ **$-3$**

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can complete a table of values for a straight-line graph | ☐ |
| SILVER | I can plot and draw straight-line graphs | ☐ |
| GOLD | I can draw lines such as $x = 3$, $y = -2$ and $x + y = 5$ | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Plotting Graphs

INTRODUCTION · 3 MIN

Make a table, plot the points, join them with a ruler.

## Table of values

Substitute each x value into the equation.

$y = 2x$: $x = 3$ gives $y = 6$

## Plot

Plot each pair (x, y) as a point.

(3, 6)

## Join

Draw one straight line through all the points.

Use a ruler

## Special lines

$x = a$ is vertical. $y = b$ is horizontal.

$x = 3$, $y = -2$

# Examples 1 and 2 — Tables of Values

EXAMPLES · 5 MIN

BRONZE

*Watch and copy into your book*

## Example 1

Complete the table of values for $y = 2x - 1$

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | | | | | |

## Example 2

Draw the lines $x = 3$ and $y = -2$ on the same axes.

# Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

*Check your copy and correct in green pen*

## Example 1

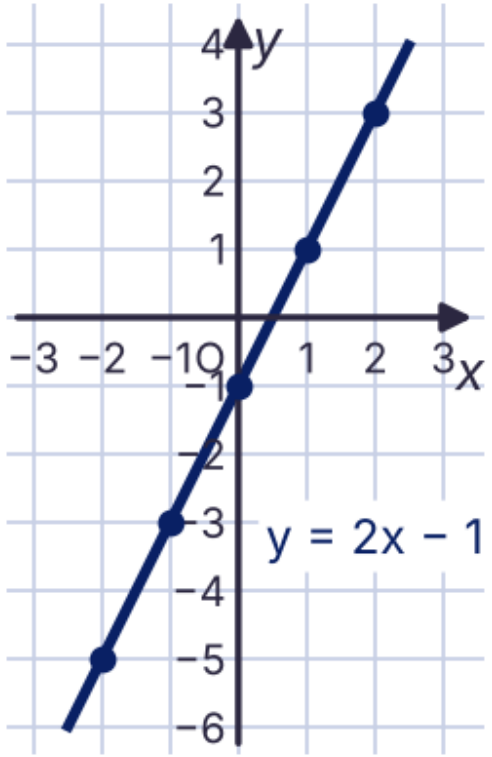
x = −2:  y = −5

x = −1:  y = −3

x = 0:  y = −1

x = 1:  y = 1

**x = 2:  y = 3**

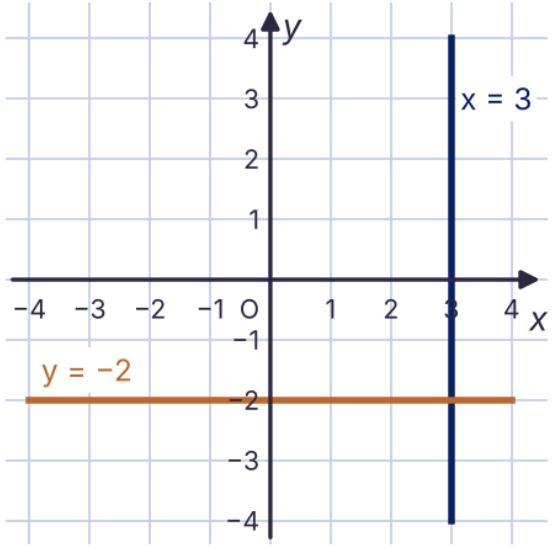



Tip: substitute every x value, then plot.

## Example 2

x = 3:  vertical

**y = −2:  horizontal**




Tip: every point on x = 3 has x-coordinate 3.

# Task 1

PRACTICE · 9 MIN

BRONZE

*Copy and complete each table of values*

**1** $y = x + 3$

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | | | | | |

**2** $y = 3x$

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | | | | | |

**3** $y = 2x + 2$

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | | | | | |

**4** $y = 4 - x$

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | | | | | |

**5** $y = 2x - 3$

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $y$ | | | | | |

★ Write the equation of the horizontal line through (0, 4)

# Task 1 — Answers

CORRECTIONS

BRONZE

*Tick or correct in green pen*

**1**

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | 1 | 2 | 3 | 4 | 5 |

**2**

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | −6 | −3 | 0 | 3 | 6 |

**3**

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | −2 | 0 | 2 | 4 | 6 |

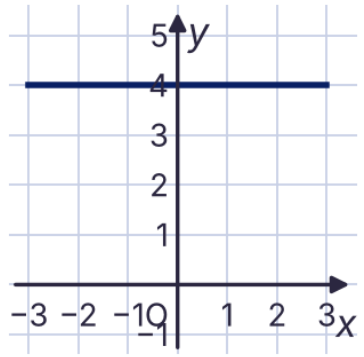
**4**

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | 6 | 5 | 4 | 3 | 2 |

**5**

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $y$ | −3 | −1 | 1 | 3 | 5 |

★ $y = 4$



# Examples 3 and 4 — Drawing Straight Lines

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

## Example 3

Draw the graph of $y = 3x - 2$

for values of x from −1 to 3

## Example 4

Draw the graph of $x + y = 5$

# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

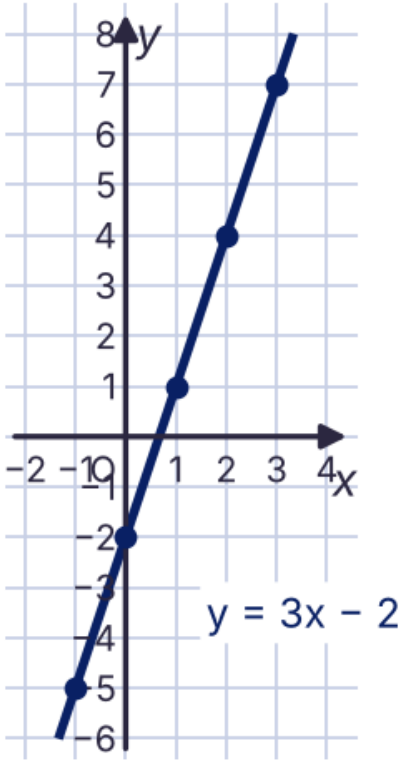
x = −1:  y = −5

x = 0:  y = −2

x = 1:  y = 1

x = 2:  y = 4

**x = 3:  y = 7**



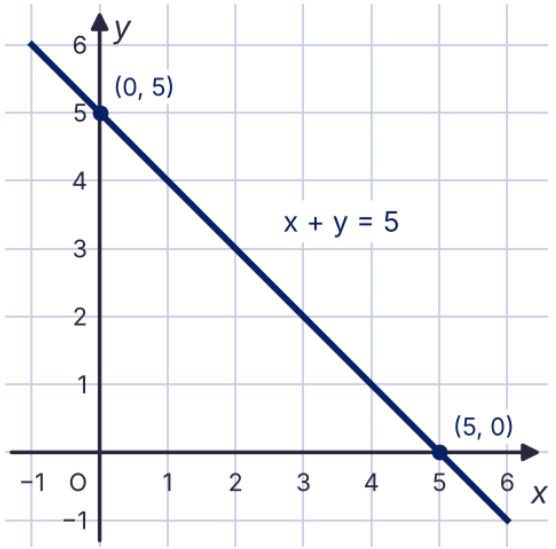

Tip: plot at least three points to check.

## Example 4

x = 0:  y = 5

y = 0:  x = 5

**(0, 5) and (5, 0)**




Tip: cover-up method: put x = 0, then y = 0.

# Task 2

PRACTICE · 9 MIN

SILVER

*Draw each graph in your book (use x from −2 to 2 for 1 and 2)*

1. $y = 2x + 3$
2. $y = -2x + 1$
3. $x + y = 6$
4. $2x + y = 8$
5. $3x + 2y = 12$

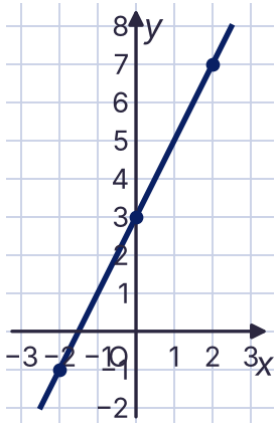
# Task 2 — Answers

CORRECTIONS

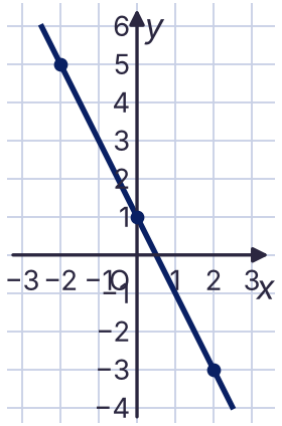
SILVER

*Tick or correct in green pen*

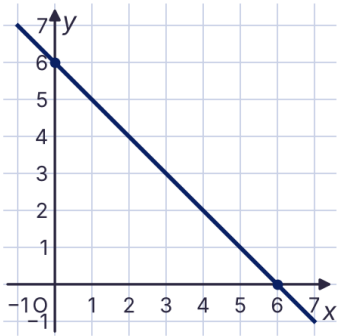
**1** $y = 2x + 3$



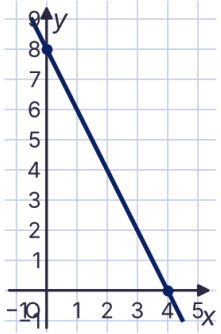

**2** $y = -2x + 1$



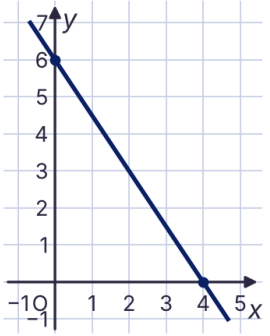

**3** (0, 6) and (6, 0)




**4** (0, 8) and (4, 0)




**5** (0, 6) and (4, 0)



# Examples 5 and 6 — Points on Lines

EXAMPLES · 5 MIN

GOLD

*Exam-style: show all your working*

## Example 5

(2 marks)

Does the point (3, 11) lie on the line $y = 4x - 1$?

Show your working.

## Example 6

(2 marks)

The point (k, 7) lies on the line $3x + 2y = 20$.

Find the value of k.

# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

$$y = 4 \times 3 - 1$$
$$= 11$$

**Yes, (3, 11) lies on the line**

Tip: substitute x and check you get y.

## Example 6

$$3k + 2 \times 7 = 20$$
$$3k + 14 = 20$$
$$3k = 6$$
$$\mathbf{k = 2}$$

Tip: substitute the coordinates, then solve.

# Task 3

PRACTICE · 9 MIN

GOLD

*Exam-style: show all your working*

1. Does the point (2, 7) lie on the line $y = 3x + 1$? Show your working. (2 marks)

2. The point $(k, -1)$ lies on the line $y = 2x - 5$. Find $k$. (2 marks)

3. Find where the line $2x + 5y = 20$ crosses each axis. (2 marks)

4. Draw $y = x + 1$ and $y = 5 - x$ on the same axes. Write down where they cross. (3 marks)

5. The point $(3, k)$ lies on the line $4x - 3y = 6$. Find $k$. (2 marks)

# Task 3 — Answers

CORRECTIONS

GOLD

*Tick or correct in green pen*

**1**

$y = 3 \times 2 + 1$

$= 7$

**Yes**

**2**

$-1 = 2k - 5$

$2k = 4$

**$k = 2$**

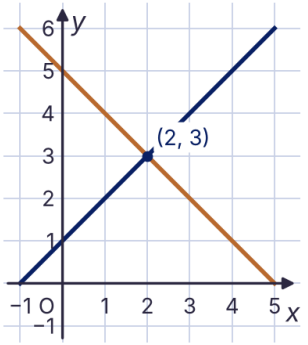
**3**

$x = 0$: $y = 4$

$y = 0$: $x = 10$

**(0, 4) and (10, 0)**

**4** They cross at (2, 3)




**5**

$12 - 3k = 6$

$3k = 6$

**$k = 2$**

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** $y = x - 2$. Find y when x = 0, 1 and 2

**SILVER** Where does $x + y = 7$ cross the axes?

**GOLD** The point (k, 4) lies on $2x + 3y = 18$. Find k.

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

**BRONZE** $y = -2, \quad y = -1, \quad y = \mathbf{0}$

**SILVER** **(0, 7) and (7, 0)**

**GOLD** $2k + 12 = 18, \quad 2k = 6, \quad k = \mathbf{3}$

# Exam Practice — Tables of Values

EXAM PRACTICE · 10 MIN

BRONZE

*IGCSE exam-style: answer on the line, show all your working*

## Question 1

(3 marks)

Here is a table of values for y = 2x + 1

| x | −1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|
| y | −1 | | 3 | | |

**(a)** Complete the table of values. (2)

.....

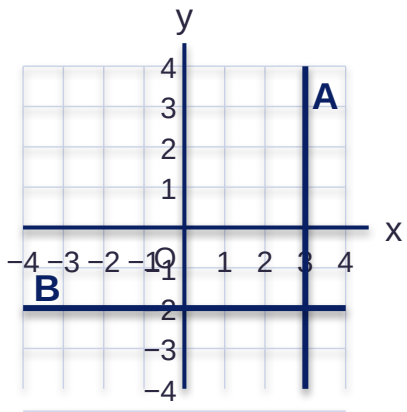
**(b)** Write down the coordinates of the point where the line crosses the y-axis. (1)

( ..... , ..... )

## Question 2

(2 marks)

The grid shows two lines, A and B.




**(a)** Find the equation of line A. (1)

.....

**(b)** Find the equation of line B. (1)

.....

# Exam Practice — Drawing Line Graphs

EXAM PRACTICE · 10 MIN

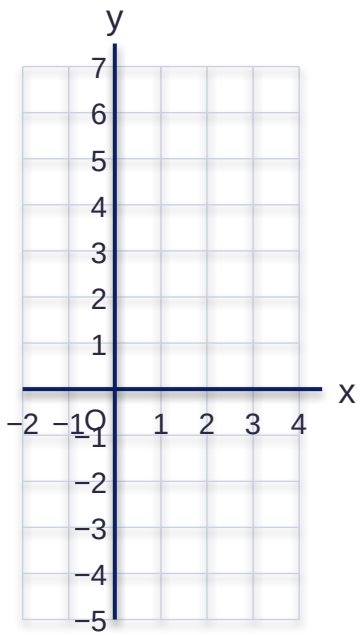
SILVER

*IGCSE exam-style: answer on the line, show all your working*

## Question 3

(3 marks)

On the grid, draw the graph of $y = 3x - 2$ for values of $x$ from $-1$ to $3$.



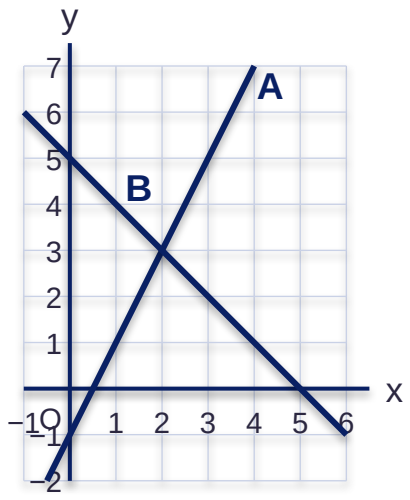

## Question 4

(1 mark)

Line A has equation $y = 2x - 1$.

Line B has equation $x + y = 5$.

Write down where A and B meet.




( ..... , ..... )

# Exam Practice — Points on a Line

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: answer on the line, show all your working*

## Question 5

(2 marks)

Does the point (4, 11) lie on the line $y = 3x - 1$?

You must show your working.

.....

## Question 6

(4 marks)

The line L has equation $2x + 3y = 21$.

**(a)** The point (k, 5) lies on L. Find the value of k. (2)

k = .....

**(b)** The point (6, m) lies on L. Find the value of m. (2)

m = .....

# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

**(a) y = 1, 5, 7** — B2

B1 for any two correct

**(b) (0, 1)** — B1

Tip: substitute each x into 2x + 1.

## Question 2

**(a) x = 3** — B1

**(b) y = −2** — B1

Tip: vertical lines are x = …, horizontal are y = …

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

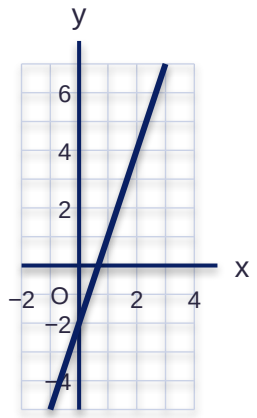
*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

y = −5, −2, 1, 4, 7 — **M1**

Plots points correctly — **M1**

**Straight line from x = −1 to 3** — **A1**




Tip: make a table of values first.

## Question 4

**(2, 3)** — **B1**

Check: 3 = 2(2) − 1 ✓

2 + 3 = 5 ✓

Tip: check the point in BOTH equations.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

$3 \times 4 - 1 = 11$ — M1

**Yes, it lies on the line** — A1

Tip: substitute x, compare with y.

## Question 6

(a) $2k + 15 = 21$ — M1

**$k = 3$** — A1

(b) $12 + 3m = 21$ — M1

**$m = 3$** — A1

Tip: substitute the known coordinate, then solve.